

Introduction to mathematics for *altx*

Wigner Summer Camp

Data and Compute Intensive Sciences Research Group

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Contents

Vectors

Dot product

Matrices

Vector-Matrix Multiplication

Matrix-Matrix Multiplication

Eigenvalues and Eigenvectors

Outline

Vectors

Dot product

Matrices

Vector-Matrix Multiplication

Matrix-Matrix Multiplication

Eigenvalues and Eigenvectors

What is a Vector?

- ▶ **Mathematician's view:** A set that satisfies the 8 axioms.
- ▶ **Physicist's view:** An arrow in space with direction and magnitude.
- ▶ **Computer Scientist's view:** A 1D array or list of numbers.
- ▶ **Notation:**

$$\vec{v} = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix}, \quad \mathbf{w} = \langle 4, 0, -5 \rangle, \quad \vec{u} = (1, 2, 3). \quad (1)$$

- ▶ **Boldface:** $\mathbf{v}$ (common in CS and mathematical books).
- ▶ **Arrow:** $\vec{v}$ (common in physics).
- ▶ **Angled brackets or parentheses.**

Example vectors

Example 1 (Vectors)

$$\vec{a} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \vec{b} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \vec{c} = \begin{bmatrix} -3 \\ 2 \\ 5 \end{bmatrix}. \quad (2)$$

Example 2 (Common vectors)

$$\vec{i} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \vec{j} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \vec{0} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \quad (3)$$

What is the Dot Product?

- ▶ The dot product of two vectors $\vec{a}$ and $\vec{b}$ is:

$$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + \cdots + a_n b_n. \quad (4)$$

- ▶ It can be shown that:

$$\vec{a} \cdot \vec{b} = \|\vec{a}\| \cdot \|\vec{b}\| \cos\left(\angle\left(\vec{a}, \vec{b}\right)\right). \quad (5)$$

Example 3 (Dot product)

$$\begin{bmatrix} 2 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ -1 \end{bmatrix} = 2 \cdot 4 + 3 \cdot (-1) = 8 - 3 = 5. \quad (6)$$

Outline

Vectors

Dot product

Matrices

Vector-Matrix Multiplication

Matrix-Matrix Multiplication

Eigenvalues and Eigenvectors

Dot Product – Practice Exercises

- Calculate the dot product of the following vector pairs:

1. $\begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 0 \\ -1 \end{bmatrix}.$

2. $\begin{bmatrix} 2 \\ 5 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$

Dot Product – Solutions

► **Solution to (a):**

$$1 \cdot 4 + (-2) \cdot 0 + 3 \cdot (-1) = 4 + 0 - 3 = \boxed{1}.$$

► **Solution to (b):**

$$2 \cdot (-1) + 5 \cdot 1 = -2 + 5 = \boxed{3}.$$

► **Interpretation:**

- (a) Zero $\Rightarrow$ the vectors are perpendicular.
- (b) Positive $\Rightarrow$ angle between the vectors is acute.

What is a Matrix?

- ▶ A 2D array of numbers arranged in rows and columns:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \quad (7)$$

where A is a 2×3 matrix.

- ▶ Types:
 - ▶ **Square:** same number of rows and columns (e.g., 2×2 , 3×3). Mathematicians like them.
 - ▶ **Non-square:** different number of rows and columns (e.g., 2×3 , 3×2).

Outline

Vectors

Dot product

Matrices

Vector-Matrix Multiplication

Matrix-Matrix Multiplication

Eigenvalues and Eigenvectors

Vector-Matrix Multiplication

- Each row of the matrix is dotted with the vector:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}, \quad \vec{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}. \quad (8)$$

- Multiplication:

$$A\vec{v} = \begin{bmatrix} 1 \cdot 2 + 2 \cdot 1 \\ 3 \cdot 2 + 4 \cdot 1 \\ 5 \cdot 2 + 6 \cdot 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 10 \\ 16 \end{bmatrix}. \quad (9)$$

Outline

Vectors

Dot product

Matrices

Vector-Matrix Multiplication

Matrix-Matrix Multiplication

Eigenvalues and Eigenvectors

Vector-Matrix Multiplication – Exercises

- Multiply the vector $\vec{v}$ with matrix M in the following examples:

1. General case ($\vec{v}M$):

$$\vec{v} = \begin{bmatrix} 1 & 2 \end{bmatrix}, \quad M = \begin{bmatrix} 3 & 0 & 1 \\ 1 & 4 & 2 \end{bmatrix}. \quad (10)$$

2. Interesting effect ($M_1\vec{v}$ and $M_2\vec{v}$):

$$\vec{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad M_1 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \quad M_2 = \begin{bmatrix} -2 & -2 \\ -2 & -2 \end{bmatrix}. \quad (11)$$

3. Even more interesting effect ($M_3\vec{v}$ and $M_4\vec{v}$):

$$\vec{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad M_3 = \begin{bmatrix} 1 & -1 \\ 2 & -2 \end{bmatrix}, \quad M_4 = \begin{bmatrix} 5 & -5 \\ -3 & 3 \end{bmatrix}. \quad (12)$$

- Try to interpret the geometric meaning of each result.

Vector-Matrix Multiplication – Solutions

1. General case ($\vec{v}M$):

$$\vec{v} \cdot M = \begin{bmatrix} 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 3 & 0 & 1 \\ 1 & 4 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 8 & 5 \end{bmatrix}. \quad (13)$$

2. Stretching cases ($M_1\vec{v}$ and $M_2\vec{v}$):

$$\vec{v} \cdot M_1 = \begin{bmatrix} 2 \\ 2 \end{bmatrix} = 2\vec{v}, \quad \vec{v} \cdot M_2 = \begin{bmatrix} -4 \\ -4 \end{bmatrix} = -4\vec{v}. \quad (14)$$

3. Nearly null output ($M_3\vec{v}$ and $M_4\vec{v}$):

$$\vec{v} \cdot M_3 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = 0\vec{v}, \quad \vec{v} \cdot M_4 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = 0\vec{v}. \quad (15)$$

Matrices as linear transformations

- ▶ Each matrix denotes a transformation $\vec{v} \mapsto M\vec{v}$. It's always linear.
- ▶ It can be proven that all linear transformations can be described with a matrix.

Transformation examples + exercise

- ▶ Stretching: $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$.
- ▶ Rotation: $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$.
- ▶ Reflection: $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$.
- ▶ Projection (flattening onto a line or plane) $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$.

Exercise: after these transformation, where do vectors:

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \text{ and } \begin{bmatrix} 1 \\ 1 \end{bmatrix}. \quad (16)$$

transformed to?

Transformation Examples – Solution 1

The stretching matrix $S = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ scales every vector by a factor of 2.

► For $\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$:

$$S\vec{e}_1 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix} = 2\vec{e}_1. \quad (17)$$

► For $\vec{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$:

$$S\vec{e}_2 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix} = 2\vec{e}_2. \quad (18)$$

► For $\vec{d} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$:

$$S\vec{d} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix} = 2\vec{d}. \quad (19)$$

Transformation Examples – Solution 2

The rotation matrix $R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ rotates every vector by 90° counterclockwise.

► For $\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$:

$$R\vec{e}_1 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \vec{e}_2. \quad (20)$$

► For $\vec{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$:

$$R\vec{e}_2 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix} = -\vec{e}_1. \quad (21)$$

► For $\vec{d} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$:

$$R\vec{d} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}. \quad (22)$$

Transformation Examples – Solution 3

The reflection matrix $F = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ flips the sign of the y -component.

► For $\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$:

$$F\vec{e}_1 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \vec{e}_1. \quad (23)$$

► For $\vec{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$:

$$F\vec{e}_2 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} = -\vec{e}_2. \quad (24)$$

► For $\vec{d} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$:

$$F\vec{d} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}. \quad (25)$$

Transformation Examples – Solution 4

The projection matrix $P = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ sets the y -component to zero.

► For $\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$:

$$P\vec{e}_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \vec{e}_1. \quad (26)$$

► For $\vec{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$:

$$P\vec{e}_2 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \vec{0}. \quad (27)$$

► For $\vec{d} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$:

$$P\vec{d} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \vec{e}_1. \quad (28)$$

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Dot product

Matrices

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Eigenvalues and Eigenvectors

Matrix-Matrix Multiplication

Procedure is the same as the Vector-Matrix multiplication. If an $n \times m$ matrix is multiplied by an $m \times k$, the result is an $n \times k$ matrix.

- ▶ The number of columns in the first matrix must be equal to the number of rows in the second matrix.

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \quad B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \quad (29)$$

$$A \cdot B = \begin{bmatrix} a_{11} \cdot b_{11} + a_{12} \cdot b_{21} & a_{11} \cdot b_{12} + a_{12} \cdot b_{22} \\ a_{21} \cdot b_{11} + a_{22} \cdot b_{21} & a_{21} \cdot b_{12} + a_{22} \cdot b_{22} \end{bmatrix} \quad (30)$$

Example 4

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix} \Rightarrow A \cdot B = \begin{bmatrix} 4 & 4 \\ 10 & 8 \end{bmatrix} \quad (31)$$

Other examples

Example 5

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \Rightarrow A \cdot B = \begin{bmatrix} 4 & 5 \\ 10 & 11 \end{bmatrix} \quad (32)$$

Example 6

$$A = \begin{bmatrix} 8 & 4 & 11 & 5 \\ 4 & 4 & 21 & 6 \\ 6 & 9 & 11 & 8 \end{bmatrix}, B = \begin{bmatrix} 3 & 2 \\ 4 & 6 \\ 1 & 4 \\ 7 & 2 \end{bmatrix} \Rightarrow A \cdot B = \begin{bmatrix} 86 & 94 \\ 91 & 128 \\ 121 & 126 \end{bmatrix} \quad (33)$$

Outline

Vectors

Dot product

Matrices

Vector-Matrix Multiplication

Matrix-Matrix Multiplication

Eigenvalues and Eigenvectors

Eigenvalues and Eigenvectors

- ▶ Notice that

$$\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} . \quad (34)$$

- ▶ Generally for a square matrix A if

$$A\vec{v} = \lambda\vec{v}, \quad (35)$$

then:

- ▶ $\vec{v}$ is an eigenvector and
- ▶ scalar λ is the corresponding eigenvalue.
- ▶ These vectors don't change direction when multiplied by A .

Eigenvalues and Eigenvectors – Properties

- ▶ If $A\vec{v} = \lambda\vec{v}$, then $\vec{v}$ is an eigenvector of A with eigenvalue λ .
- ▶ Eigenvectors are defined up to a scalar: if $\vec{v}$ is an eigenvector, so is $c\vec{v}$ for any $c \neq 0$.
- ▶ The set of all eigenvectors corresponding to a single eigenvalue forms a vector subspace.
- ▶ A square matrix of size $n \times n$ has at most n eigenvalues (including complex ones and multiplicities).
- ▶ If all eigenvalues of a matrix are positive, the matrix is positive definite.
- ▶ Diagonal matrices (matrices where for all $a_{ij} = 0$ where $i \neq j$) have their diagonal elements as eigenvalues, and the standard basis vectors as eigenvectors.
- ▶ The eigenvalues of a complex Hermitian or real symmetric matrix are real.

Eigenvalues with PyTorch

- ▶ `torch.linalg.eigh()` computes eigenvalues and orthonormal eigenvectors of a real symmetric matrix.
- ▶ Documentation: `torch.linalg.eigh`.
- ▶ It returns a tuple (`eigenvalues`, `eigenvectors`), where:
 - ▶ `eigenvalues` is a 1D tensor sorted in ascending order,
 - ▶ each column of `eigenvectors` is the corresponding unit-length eigenvector.

```
1 import torch
2
3 A = torch.tensor([[2., 1.],
4                  [1., 2.]])
5 eigenvalues, eigenvectors = torch.linalg.eigh(A)
6 # eigenvalues:  tensor([1., 3.])
7 # eigenvectors: tensor([[ -0.7071,  0.7071],
8 #                      [ 0.7071,  0.7071]])
```

Scaling Eigenvectors to Integer Form

- ▶ `torch.linalg.eigh()` always returns *unit* eigenvectors, i.e. eigenvectors with norm 1.
- ▶ The integer eigenvectors of $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ are $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$, each with norm $\sqrt{2}$.
- ▶ Multiply the unit eigenvectors by $\sqrt{2}$ to recover the integer form.

```
1 integer_eigvecs = eigenvectors * np.sqrt(2)
2 # tensor([[ -1.,   1.],
3 #         [ 1.,   1.]]) (each column is one
   eigenvector)
```

- ▶ In general, to recover an integer eigenvector $\vec{w}$, multiply the corresponding unit eigenvector by $\|\vec{w}\|$.

PyTorch Eigendecomposition – Exercise

Use `torch.linalg.eigh()` to find the eigenvalues and eigenvectors of

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}. \quad (36)$$

- ▶ What are the eigenvalues?
- ▶ Scale the returned unit eigenvectors to their integer form using the technique from the previous slide.
- ▶ What are the integer eigenvectors?

PyTorch Eigendecomposition – Solution (1/2)

```
1 import torch
2
3 A = torch.tensor([[3., 1.],
4                   [1., 3.]])
5 eigenvalues, eigenvectors = torch.linalg.eigh(A)
6 # eigenvalues: tensor([2., 4.])
7 # eigenvectors: tensor([[ -0.7071,  0.7071],
8 #                       [ 0.7071,  0.7071]])
9
10 integer_eigvecs = eigenvectors * np.sqrt(2)
11 # tensor([[ -1.,  1.],
12 #         [ 1.,  1.]])
```

PyTorch Eigendecomposition – Solution (2/2)

- ▶ The eigenvalues are $\lambda_1 = 2$ and $\lambda_2 = 4$.
- ▶ The integer eigenvectors, obtained by multiplying the unit eigenvectors by $\sqrt{2}$, are:

$$\vec{v}_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \text{ for } \lambda_1 = 2, \quad \vec{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ for } \lambda_2 = 4. \quad (37)$$

- ▶ Verification of the eigenvalue equation $A\vec{v} = \lambda\vec{v}$:

$$\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \end{bmatrix} = 2 \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 1 \end{bmatrix}. \quad (38)$$

Revision

In this presentation you learned about:

- ▶ Vectors.
- ▶ Matrices.
- ▶ Dot products.
- ▶ Vector-Matrix and Matrix-Matrix multiplication.
- ▶ Eigendecomposition.
- ▶ Computing eigenvalues and eigenvectors with PyTorch.